

GREEN BUILDINGS: Architectures For A Green And Sustainable India

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According to World Green Building Council, a green building is a building that, in its design, construction or operation, reduces or eliminates negative impacts, and can create positive impacts, on our climate and natural environment. A green building preserves precious natural resources and improves our quality of life

"Green buildings are a hallmark of economically sound business decisions, thoughtful environmental decisions and smart human impact decisions."

—Rick Fedrizzi, chairman, International WELL Building Institute

The present environmental conditions of major Indian cities require immediate attention. One root cause lies in most residential and commercial buildings that consume much higher resources than needed, such as water, electricity and fuel. Embracing green buildings can be a smart move forwards, as these have the capacity to preserve such resources, cut down pollutants and ensure long-term paybacks.

The benefits

Right from the construction phase of conventional buildings, there is massive wastage of raw materials. It leaves a great trace of dry waste that adds to air, water and soil to pollute these. Building construction accounts for up to 40 per cent greenhouse gas emissions. Occupied households, on the other hand, utilise electricity and water at a high rate for daily use.

In addition, use of equipment, like refrigerators and air-conditioners, adds up to another 18 per cent greenhouse gases. Other forms of waste

(wastewater, solid, biodegradable wastes, etc) also factor in heavily.

On the other hand, the construction phase of green buildings is well planned, to reduce the wastage of material and resources. Green buildings can save at least 20 per cent of the materials used. In addition, many of the materials used for green building constructions can either be recycled or biodegraded, like hempcrete, bamboo, wood, straw bales, recycled plastic and so on. Lesser construction waste improves air quality and reduces carbon emissions.

Ashu Dehadani, manager, Green Rating for Integrated Habitat Assessment (GRIHA) Council, says, "Through the use of various energy-efficient methods, at least 15 per cent of electricity can be saved. This is a major financial saving at the end of the year for a person."

According to Indian Green Building Council (IGBC), "Green buildings are capable of reducing 30 to 50 per cent water consumption. Further intangible benefits include highly improved indoor air quality, optimal daylighting, safety and an overall improved, healthy way of living for occupants."

Dehadani adds, "Green buildings are more weather-resistant. Be it heavy monsoon, summer or winter, green constructions keep indoors and the whole setup well-regulated and damage-free. In many setups, these can even utilise weather conditions for further benefits. For example, excessive rainfall can be used for an occupant's benefit through rainwater harvesting.

"Wastes are segregated at source. Biodegradable wastes are transported for bio-gas and manure creation, while non-biodegradable ones are recycled and reused. In these ways, green buildings deliver many essential benefits."

What constitutes green buildings

A green construction is the result of an overall design approach that addresses various environmental, financial and aesthetic parameters, holistically. More importantly, it addresses the comfort and well-being of the occupants of the building without pulling any additional investments going forwards.

CII Green Business Centre (Source: www.greenbusinesscentre.com)

